mass on the end of a spring. But it's important not to get confused here: the oscillators are not masses attached to springs. What's really oscillating are the electric and magnetic fields. For each wavelength, there is a mathematical harmonic oscillator describing the amplitude or strength of the field. That's a lot of harmonic oscillators all running simultaneously. Fortunately, however, they all oscillate independently, so we can focus our attention on waves of one particular wavelength and ignore all the others.

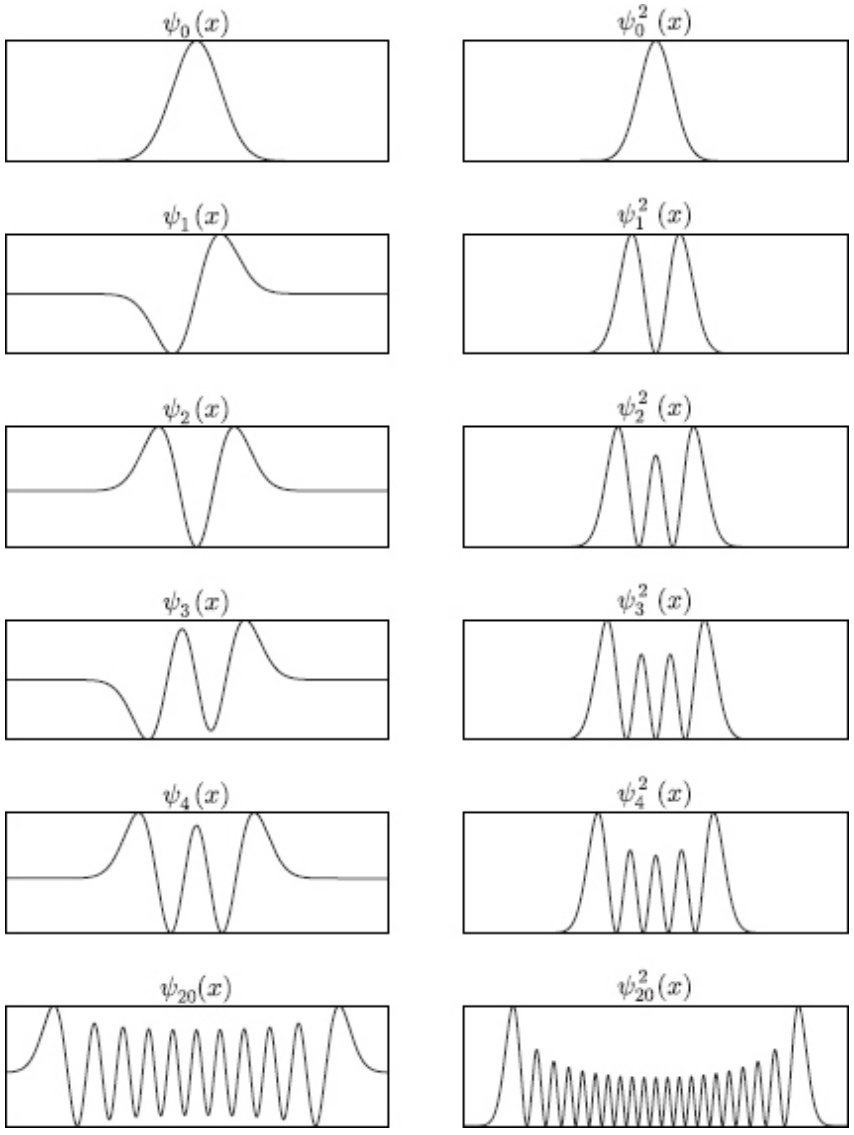


Figure 10.3: Harmonic Oscillator Eigenfunctions. Amplitudes are shown on the left, probabilities on the right. The higher-energy wave functions oscillate more rapidly and are more spread out.